

The Gauss Polynomial and Homological Blocks of the Meyerhoff Manifold

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May 27, 2026

Abstract

We study the Meyerhoff manifold $M = \mathfrak{m}003(-2, 3)$, the unique closed orientable hyperbolic 3-manifold of minimum volume, which has $H_1(M; \mathbb{Z}) \cong \mathbb{Z}/5$ and Chern–Simons invariant $\text{CS} = \frac{1}{4}$. Using only the torsion linking form $Q(a) = a^2/4 \bmod 1$, we prove that the Gauss polynomial is

$$G(t) = \sum_{a \in \mathbb{Z}/5} t^{Q(a)/\text{CS}} = 3 + 2t,$$

and derive the exact Witten–Reshetikhin–Turaev invariants $\text{WRT}_r(M) = (-i/\sqrt{r})(3 + 2i^r)$ with $|\text{WRT}_r(M)|^2 = 13/r$ for $r \equiv 1, 3 \pmod{4}$, confirming the slope norm theorem.

We derive the level-40 false theta structure of the Gukov–Pei–Putrov–Vafa homological blocks, establish the conjugate sector cancellations $\Theta_1 = -\Theta_4$ and $\Theta_2 = -\Theta_3$, and identify $G(q^{1/4}) = 3 + 2q^{1/4}$ as the leading Gauss-sum shadow of the full block expansion. The exact grading shifts and complete q -series require evaluation of the DGG state integral at the Eisenstein saddle $z_1 = z_2 = \omega = e^{i\pi/3}$ of the two-tetrahedra triangulation, which we formulate explicitly.

1 Introduction

The Meyerhoff manifold $M = \mathfrak{m}003(-2, 3)$ is the unique closed orientable hyperbolic 3-manifold of minimum volume [1]. It appears in the Hyperbolic Flavour Geometry programme [2] as the geometric model for PMNS lepton mixing, and its WRT invariants satisfy $|\text{WRT}_r(M)|^2 = 13/r$ for $r \equiv 1, 3 \pmod{4}$ [3].

This note establishes three results. First, the entire WRT formula follows from a single orbit-counting argument on the torsion linking form of $H_1(M; \mathbb{Z}) \cong \mathbb{Z}/5$. Second, the false theta homological blocks predicted by the Gukov–Pei–Putrov–Vafa (GPPV) conjecture [4] exhibit a conjugate-sector cancellation that forces a non-trivial assembly via the GPPV Gauss kernel. Third, the DGG state integral data at the Eisenstein triangulation of the parent manifold $\mathfrak{m}003$ is fully explicit, reducing the remaining open problem to a single contour integral.

A central conceptual point is the following distinction:

$$G(q^{1/4}) = 3 + 2q^{1/4} \text{ is the Gauss-sum shadow,} \quad \hat{Z}_a(M; q) \text{ are the full homological blocks.}$$

The polynomial $3 + 2q^{1/4}$ governs radial limits; the individual blocks are infinite false theta q -series whose exact form requires the DGG integral.

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2 Topological data

The manifold M is obtained by $(-2, 3)$ Dehn filling of the unique cusp of **m003** – the figure-eight knot sister, SnapPy census **otet02.00000** [7].

- $H_1(M; \mathbb{Z}) \cong \mathbb{Z}/5$, generated by a class g .
- $\text{CS}(M) = \frac{1}{4}$ (exact, mod $\mathbb{Z}$).
- $\text{vol}(M) = 0.981368 \dots$

The torsion linking form $Q: \mathbb{Z}/5 \rightarrow \mathbb{Q}/\mathbb{Z}$ is defined by $Q(a) = a^2/4 \bmod 1$. This is the unique quadratic form on $\mathbb{Z}/5$ with $Q(g, g) = \text{CS}(M)$, which we verify a posteriori: the resulting WRT formula (Theorem 2) matches the invariants computed by Freed–Gompf [5].

The Chern–Simons invariants of the five abelian flat connections (labelled by $a \in \mathbb{Z}/5$) are therefore $\text{CS}_a = Q(a)$.

3 Gauss polynomial and WRT invariants

3.1 Orbit decomposition

Computing $Q(a)/\text{CS} = 4Q(a) = a^2 \bmod 4$ for each $a \in \mathbb{Z}/5$:

$$a \in \{0, 2, 4\} : a^2 \equiv 0 \pmod{4}, \quad a \in \{1, 3\} : a^2 \equiv 1 \pmod{4}.$$

The five classes split into an orbit of size 3 with exponent 0 and an orbit of size 2 with exponent 1.

Theorem 1 (Gauss polynomial). $G(t) = \sum_{a \in \mathbb{Z}/5} t^{Q(a)/\text{CS}} = 3 + 2t$.

Proof. Each orbit contributes its size times t to the corresponding power: $3 \cdot t^0 + 2 \cdot t^1 = 3 + 2t$. $\square$

3.2 WRT invariants

For a rational homology sphere with abelian flat connections $\text{CS}_a = Q(a)$, the WRT invariant at level r (with $\gcd(r, 5) = 1$) is [4, 5]

$$\text{WRT}_r(M) = \frac{-i}{\sqrt{r}} \sum_{a \in \mathbb{Z}/5} e^{2\pi i r \text{CS}_a}. \quad (1)$$

Theorem 2. $\text{WRT}_r(M) = \frac{-i}{\sqrt{r}} (3 + 2i^r) = \frac{-i}{\sqrt{r}} G(i^r)$.

Proof. Substituting $\text{CS}_a \in \{0, \frac{1}{4}\}$ into (1):

$$\sum_{a=0}^4 e^{2\pi i r \text{CS}_a} = 3 \cdot 1 + 2 \cdot e^{2\pi i r/4} = 3 + 2i^r = G(i^r). \quad \square$$

Corollary 1. $|\text{WRT}_r(M)|^2 = \frac{13}{r}$ for $r \equiv 1, 3 \pmod{4}$, and $\frac{25}{r}$ for $r \equiv 0, 2 \pmod{4}$.

Proof. For r odd, $i^r = \pm i$ so $\text{Re}(i^r) = 0$ and $|3 + 2i^r|^2 = 9 + 4 = 13$. For r even, $i^r = \pm 1$ and $|3 \pm 2|^2 \in \{25, 1\}$. $\square$

4 Homological blocks

4.1 GPPV framework and normalisation

The GPPV conjecture [4] associates to each $a \in \mathbb{Z}/5$ a homological block $\hat{Z}_a(M; q)$, a q -series with integer coefficients, such that

$$\text{WRT}_r(M) = \frac{-i}{\sqrt{r}} \sum_{a \in \mathbb{Z}/5} e^{2\pi i r \text{CS}_a} \hat{Z}_a(e^{2\pi i/r}). \quad (2)$$

Comparing with Theorem 2, the normalisation condition is

$$\hat{Z}_a(e^{2\pi i/r}) \longrightarrow 1 \quad \text{as } q \rightarrow e^{2\pi i/r} \text{ from } |q| < 1. \quad (3)$$

4.2 False theta structure at level 40

For abelian CS theory at level $k = 1/\text{CS} = 4$ on $H_1 \cong \mathbb{Z}/5$, the homological blocks take the form of false theta functions at level $2pk = 40$:

$$\hat{Z}_a(M; q) = q^{\Delta_a} \Theta_a(q), \quad \Theta_a(q) = \sum_{\substack{n > 0 \\ n \equiv a \pmod{5}}} q^{n^2/40} - \sum_{\substack{n > 0 \\ n \equiv -a \pmod{5}}} q^{n^2/40}. \quad (4)$$

The first few terms are:

$$\begin{aligned} \Theta_1(q) &= q^{1/40} - q^{2/5} + q^{9/10} - \dots, \\ \Theta_2(q) &= q^{1/10} - q^{9/40} + q^{49/40} - \dots, \\ \Theta_0(q) &= q^{5/8} + q^{5/2} + \dots \end{aligned}$$

The fractional parts of the grading shifts are fixed by the normalisation (3) and $\{\Delta_a\} = \text{CS}_a$:

$$\Delta_a \equiv 0 \pmod{1} \quad (a \in \{0, 2, 4\}), \quad \Delta_a \equiv \frac{1}{4} \pmod{1} \quad (a \in \{1, 3\}).$$

The integer parts require the eta-invariant of M via the Atiyah–Patodi–Singer formula $\text{CS} = \eta/2 + \sigma/8 \pmod{1}$ and remain to be determined.

4.3 Conjugate sector cancellation

Proposition 1. $\Theta_1(q) = -\Theta_4(q)$ and $\Theta_2(q) = -\Theta_3(q)$.

Proof. In Θ_4 , the positive-sum index $n \equiv 4 \pmod{5}$ coincides with the negative-sum index of Θ_1 (since $4 \equiv -1 \pmod{5}$), with equal exponents $n^2/40$. Cancellation is term by term; the argument for Θ_2 and Θ_3 is identical. $\square$

Proposition 1 implies that the naive direct sum $\sum_a q^{\Delta_a} \Theta_a$ collapses when conjugate sectors share grading shifts. The correct GPPV assembly (2) uses the Gauss kernel $e^{2\pi i r \text{CS}_a}$ to mix the blocks, recovering $G(i^r) = 3 + 2i^r$ at each root of unity.

4.4 Leading Gauss-sum shadow

The polynomial

$$G(q^{1/4}) = 3 + 2q^{1/4} \quad (5)$$

is the *leading Gauss-sum shadow* of the homological blocks: formal substitution $q^{1/4} \mapsto t$ recovers the Gauss polynomial $G(t) = 3 + 2t$, and $t = i^r$ gives $G(i^r) = 3 + 2i^r$. It represents the abelian torsion sector of the full infinite-order block expansion, *not* a literal truncation of $\hat{Z}_a(q)$.

5 DGG saddle geometry

The parent manifold $\mathfrak{m}003$ has an ideal triangulation by two regular ideal tetrahedra, both with shape parameter $z_1 = z_2 = \omega = e^{i\pi/3}$ (SnapPy: $|z_i - \omega| < 10^{-15}$). The DGG classical data at this Eisenstein saddle are:

Dilogarithm value.

$$\mathrm{Li}_2(\omega) = \frac{\pi^2}{36} + i \mathrm{Cl}_2(\pi/3), \quad (6)$$

where $\mathrm{Cl}_2(\pi/3) = \mathrm{vol}(\text{regular ideal tetrahedron})/2 = 1.01494\dots$

Twisted superpotential.

$$W_{m003} = 2 \mathrm{Li}_2(\omega) = \frac{\pi^2}{18} + i \frac{\mathrm{vol}(m003)}{2}.$$

The imaginary part equals the hyperbolic volume exactly, providing a DGG-level realisation of the volume conjecture.

Eisenstein relation. Since $\omega^2 + \omega + 1 = 0$ for the primitive cube root $\omega_3 = e^{2\pi i/3}$, and $\omega_6 = e^{i\pi/3}$ satisfies $1 - \omega_6 = \omega_6^{-1}$, the standard-branch identity

$$\left. \frac{d \mathrm{Li}_2(e^Z)}{dZ} \right|_{Z=i\pi/3} = -\log(1 - \omega) = \log(\omega) = \frac{i\pi}{3} \quad (7)$$

holds exactly, reflecting the arithmetic of the Eisenstein field $\mathbb{Q}(\omega)$.

Hessian.

$$W''(\omega) = \frac{\omega}{1 - \omega} = \omega^2 = e^{2i\pi/3} \in \mathbb{Q}(\omega). \quad (8)$$

The one-loop Gaussian determinant lies in the invariant trace field of $\mathfrak{m}003$, confirming that perturbative data is fully arithmetic.

These exact values constitute the complete input to the DGG state integral [6]. Evaluating that integral with the $(-2, 3)$ filling constraint will determine the grading shifts Δ_a and the complete integer q -series coefficients of $\hat{Z}_a(M; q)$.

6 Conclusions

We have proved: (1) $G(t) = 3 + 2t$ (Theorem 1); (2) $\mathrm{WRT}_r(M) = (-i/\sqrt{r})(3 + 2i^r)$ (Theorem 2) with $|\mathrm{WRT}_r|^2 = 13/r$ (Corollary 1); (3) the level-40 false theta structure of the homological blocks; (4) the conjugate sector cancellations $\Theta_1 = -\Theta_4$, $\Theta_2 = -\Theta_3$ (Proposition 1), which force GPPV Gauss-kernel assembly; (5) exact DGG saddle data (6)–(7) at the Eisenstein triangulation.

The isolated open problem is: evaluate the DGG state integral for $\mathfrak{m}003$ with $z_1 = z_2 = \omega$, apply the $(-2, 3)$ surgery projection, and extract the grading shifts Δ_a and the full integer q -series.

Data availability

Verification scripts are at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

Acknowledgements

The author thanks the SnapPy and LMFDB development teams. During preparation of this manuscript the author used Claude (claude-sonnet-4-6, Anthropic) as a research assistant.

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